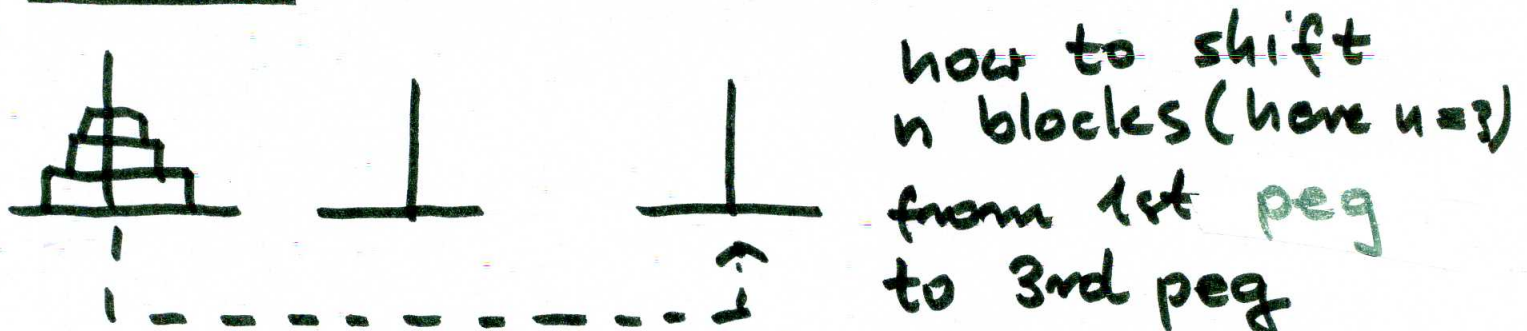


ALGORITHMS & COMPLEXITY

LECTURE 5

We shall: analyze how to transform recursive formula (from implicit form) to explicit formulation so that we can assess $a_n = O(f(n))$

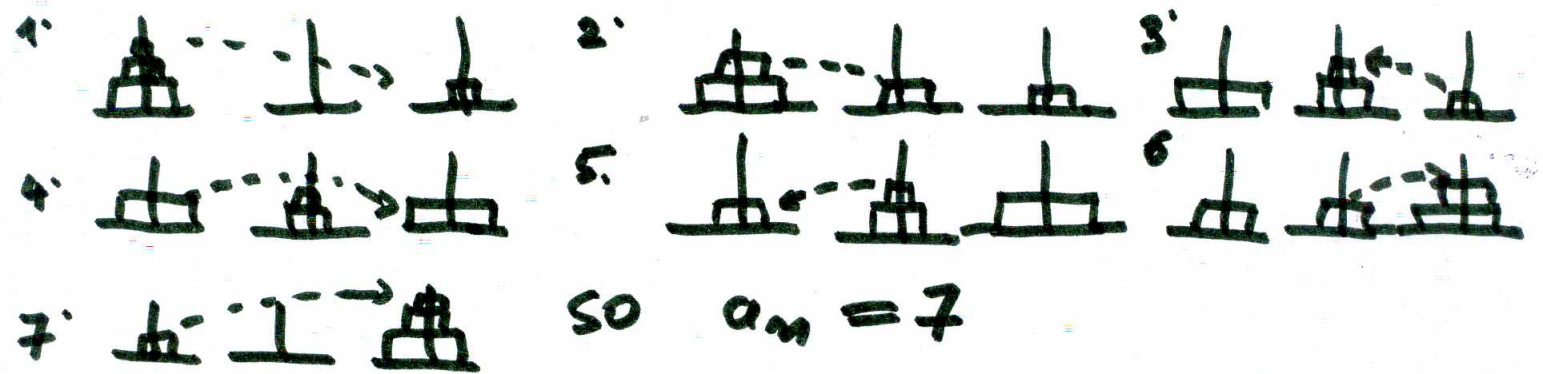
Example: Hanoi tower problem



with the aid of peg 2 subject to the constraint: "only smaller block can be put on the larger one"

a_n — the number of shifts

$$a_n = O(?)$$

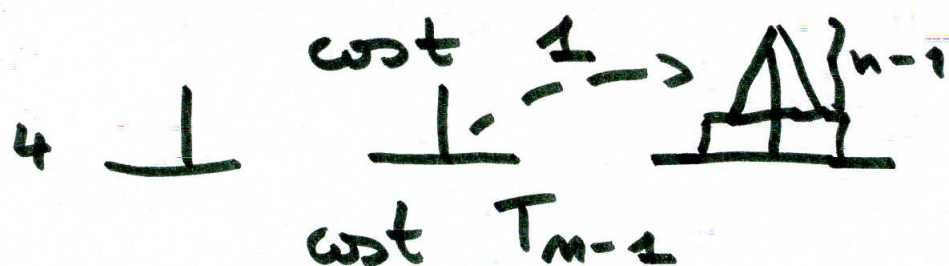
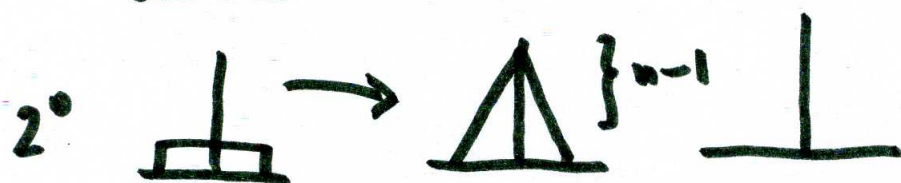
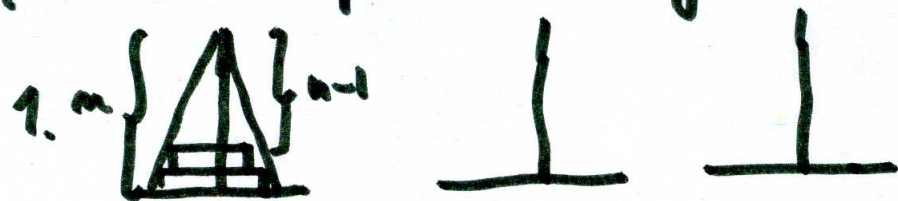


But for arbitrary n $a_n = ?$

ALGORITHMS & COMPLEXITY

LECTURES

We try to find the recursive formula
(for the specific algorithm)



Total cost $T_n = T_{n-1} + 1 + T_{n-1}$

$$\boxed{T_n = 2T_{n-1} + 1} \quad (*)$$

Perhaps now having got (*) & general theory we can transform

$$T_n = 2T_{n-1} + 1 \Rightarrow T_n = \text{explicit formula}$$

which enables to assess $T_n = O(f(n))$

ALGORITHMS & COMPLEXITY

LECTURES

Def The degree of difference equation:
is the maximal difference between
the smallest & the biggest coefficient
index in:

$$a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k}, n) \quad (*)$$

(i)

here it is k .

$$\text{If } a_n = \sum_{i=1}^k f_i(n) a_{n-i} + g(n) \quad (A)$$

(ii)

(i.e. special case of $(*)$)

then such scheme (A) is called
a linear difference scheme inhomogeneous
(as $g(n) \neq 0$) of order k .
(iii)

If $g(n) = 0$ then it is homogeneous.

If $f_i(n) = \alpha_i \leftarrow$ constants then (iv)

$$a_n = \sum_{i=1}^k \alpha_i a_{n-i} + g(n)$$

is called a linear difference scheme
of order k with constant coefficients
inhomogeneous (as $g(n) \neq 0$).
(iv) IH

If $g(n) = 0$
homogeneous
(v) H.

ALGORITHMS & COMPLEXITY

LECTURE 5

Examples:

$$1) E_n = E_{n-1} \frac{n-1}{n} + 1$$

$k=1$ IH

Linear

non-constant coeff.

$$2) a_{n+1} = \frac{1}{2} \left(a_n + \frac{4}{a_n} \right)$$

$k=1$

non-linear

$$3) a_n = a_{n-1} \cdot a_{n-2}$$

$k=2$

non-linear

$$4) a_n = 2a_{n-1} + 1$$

$k=1$, linear, IH
with constant coeff.

$$5) a_n = a_{n-10}$$

$k=10$, linear

IH, with constant
coeff. $\alpha_{10} = 1$

$$6) a_n = A \cdot T(L^{\frac{n}{2}})$$

$\alpha_1 = \dots = \alpha_9 = 0$

$k=\infty$ so this is an example
of a scheme of not fixed
order (it is DIVIDE & CONQUER)
Scheme (H)

7)

$$a_n = 1 + a_{\lfloor \frac{n}{2} \rfloor} \quad k=\infty \quad (IH)$$

$$8) a_n = a_{n-1} + \underbrace{n^2 - 2n^2 + 3n + 2}_{g(n)}$$

$k=1$, lin. $g(n)$ IH, const. coeff

$$9) a_{n+2} = a_{n+1} + a_n \quad k=2, \text{ linear, constant coeff.}$$

homogeneous

ALGORITHMS & COMPLEXITY

Usually

$k = \text{order}$

$$a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k}, n) \quad (*)$$

is supplemented with k initial values

$$a_0, a_1, \dots, a_{k-1} \quad (**)$$

in order to enforce uniqueness of a_n satisfying $(*)$ & $(**)$

Th Let $f: \mathbb{R}^k \times \mathbb{N} \rightarrow \mathbb{R}$ be a function

Then there exists exactly one sequence $\{a_n\}_{n=0}^{\infty}$ satisfying

$$a) \quad a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k}, n)$$

$$b) \quad a_i = c_i \quad 0 \leq i \leq k-1$$

$P(n)$: There exists exactly one $\{a_n\}_{n \geq 0}$ satisfying a) & b).

I) $P(0)$: true as only one sequence of numbers $a_0, a_1, \dots, a_{k-1}$ satisfies initial conditions.
 $P(1)$: true
 $\vdots$
 $P(k-1)$: true

II) Assume $P(j)$ holds $\forall j < n$
 $\Downarrow$
is $P(n)$ is true ?

ALGORITHMS & COMPLEXITY

LECTURE 5

Based on $P(n-1)$ true as $n-1 < n$
 $P(n-2)$ true as $n-2 < n$
 $\vdots$
 $P(n-k)$ true as $n-k < n$

We have only one $a_{n-1}, a_{n-2}, \dots, a_{n-k}$

Since

$$a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k}, n)$$

↑

this is uniquely determined by f
 & unique $a_{n-1}, a_{n-2}, \dots, a_{n-k} \in \mathbb{R}$

This shows $P(n)$.

By the principle of M.I. $\{a_n\}_{n \geq 0}$
 is determined uniquely.

We develop now the general approach
 to solve :

$$a_n = \sum_{i=1}^k d_i a_{n-i}$$

Fact 1: $V_\infty = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} : a_i \in \mathbb{R} \right\}$ the space of
 vectors with
 ∞ -many coefficients

ALGORITHMS & COMPLEXITY

LECTURE 3

Assume $\{a_n\}_{n \geq 0} \in V_\infty$ $\{b_n\}_{n \geq 0} \in V_\infty$

Define (i) $\{a_n\}_{n \geq 0} \oplus \{b_n\}_{n \geq 0} = \{c_n\}_{n \geq 0} \in V_\infty$

where $c_n = a_n + b_n$

(ii) $\lambda \odot \{a_n\}_{n \geq 0} = \{d_n\}_{n \geq 0} \in V_\infty$

$$d_n = \lambda \cdot a_n$$

It is easy to show that

$(V_\infty, \oplus, \odot, \mathbb{R})$ defines a vector space over a field $K = \mathbb{R}$.

Take a subset $V_\infty^K \subseteq V_\infty$

$$V_\infty^K = \left\{ \{a_n\}_{n \geq 0} : a_n = \sum_{i=1}^K d_i a_{n-i} \right\} \quad (*)$$

↑ the set of all sequences from V_∞ which satisfy the recursive scheme $a_n = \sum_{i=1}^K d_i a_{n-i}$.

V_∞^K is a linear subspace of V_∞ :

$$1) \{a_n\}_{n \geq 0}, \{b_n\}_{n \geq 0} \in V_\infty^K \Rightarrow \{a_n\}_{n \geq 0} \oplus \{b_n\}_{n \geq 0} \in V_\infty^K$$

$$2) \{a_n\}_{n \geq 0} \in V_\infty^K \Rightarrow \lambda \odot \{a_n\}_{n \geq 0} \in V_\infty^K$$

ALGORITHMS & COMPLEXITY

1) since $\{a_n\}_{n \geq 0} \in V_\infty^K$ & $\{b_n\}_{n \geq 0} \in V_\infty^K$

$$a_n = \sum_{i=1}^K d_i a_{n-i} \quad \& \quad b_n = \sum_{i=1}^K d_i b_{n-i}$$

$$\underbrace{a_n + b_n}_{c_n} = \sum_{i=1}^K d_i a_{n-i} + \sum_{i=1}^K d_i b_{n-i}$$

$$= \sum_{i=1}^K d_i \underbrace{(a_{n-i} + b_{n-i})}_{c_{n-i}}$$

$\Downarrow \{c_n\}_{n \geq 0}$ also satisfies (*)

$$\text{so } \{c_n\}_{n \geq 0} = \{a_n\}_{n \geq 0} \oplus \{b_n\}_{n \geq 0}$$

also $\bigcap_{V_\infty^K}$

$$2) \lambda \odot \{a_n\}_{n \geq 0} = \{c_n\}_{n \geq 0}$$

$$a_n = \sum_{i=1}^K d_i a_{n-i} \quad | \cdot \lambda$$

$$\underbrace{\lambda a_n}_{c_n} = \sum_{i=1}^K d_i \underbrace{\lambda a_{n-i}}_{c_{n-i}}$$

$$\text{so } \{c_n\}_{n \geq 0} = \lambda \odot \{a_n\}_{n \geq 0}$$

also satisfies (*) $\Rightarrow \{c_n\}_{n \geq 0} \in V_\infty^K$

Algorithms & complexity

LECTURE 5

Mapping:

$$\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{k-1} \end{pmatrix} \rightarrow \{a_n\}_{n \geq 0} \in V_\infty^k$$

uniquely determined

is linear

(*)

$$\begin{cases} a_0 \\ a_1 \\ \vdots \\ a_{k-1} \\ a_k = d_1 a_{k-1} + d_2 a_{k-2} + \dots + d_k a_0 \\ a_{k+1} = d_1 a_k + d_2 a_{k-1} + \dots + d_k a_1 \\ \vdots \\ a_n = d_1 a_{n-1} + d_2 a_{n-2} + \dots + d_k a_{n-k} \\ \vdots \end{cases}$$

Indeed

$$\begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_{k-1} \end{pmatrix} \rightarrow \{b_n\}_{n \geq 0} \in V_\infty^k$$

$$\varphi\left(\begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix}\right) = \{a_n\}_{n \geq 0}$$

$$\varphi\left(\begin{pmatrix} b_0 \\ \vdots \\ b_{k-1} \end{pmatrix}\right) = \{b_n\}_{n \geq 0}$$

(**)

$$\begin{cases} b_0 \\ b_1 \\ \vdots \\ b_{k-1} \\ b_k = d_1 b_{k-1} + d_2 b_{k-2} + \dots + d_k b_0 \\ b_{k+1} = d_1 b_k + d_2 b_{k-1} + \dots + d_k b_1 \\ \vdots \\ b_n = d_1 b_{n-1} + d_2 b_{n-2} + \dots + d_k b_{n-k} \\ \vdots \end{cases}$$

but $\varphi\left(\begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix}\right) + \varphi\left(\begin{pmatrix} b_0 \\ \vdots \\ b_{k-1} \end{pmatrix}\right) = \{c_n\}_{n \geq 0}$ (*) + (**)

$$\begin{cases} a_0 + b_0 \\ a_1 + b_1 \\ \vdots \\ a_{k-1} + b_{k-1} \\ a_k + b_k = d_1(a_{k-1} + b_{k-1}) + \dots + d_k(a_0 + b_0) \\ a_{k+1} + b_{k+1} = d_1(a_k + b_k) + \dots + d_k(a_1 + b_1) \\ \vdots \\ (a_n + b_n) = d_1(a_{n-1} + b_{n-1}) + \dots + d_k(a_{n-k} + b_{n-k}) \\ \vdots \end{cases}$$

(***)

but

$$\varphi\left(\begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix} \oplus \begin{pmatrix} b_0 \\ \vdots \\ b_{k-1} \end{pmatrix}\right) = \varphi\left(\begin{pmatrix} a_0 + b_0 \\ \vdots \\ a_{k-1} + b_{k-1} \end{pmatrix}\right) = ***$$

ALGORITHM & COMPLEXITY

LECTURE 5

So
$$\varphi\left(\begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix}\right) \oplus \varphi\left(\begin{pmatrix} b_0 \\ \vdots \\ b_{k-1} \end{pmatrix}\right) = \varphi\left(\begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix} + \begin{pmatrix} b_0 \\ \vdots \\ b_{k-1} \end{pmatrix}\right)$$

Similarly it is easy to show that

$$\lambda \odot \varphi\left(\begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix}\right) = \varphi\left(\lambda \cdot \begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix}\right)$$

Thus $\varphi: V^k \longrightarrow V_\infty^k$ φ is linear

The space $V^k = \left\{ \begin{pmatrix} a_0 \\ \vdots \\ a_{k-1} \end{pmatrix} : a_i \in \mathbb{R} \right\}$

It is also injective as

$(a_0, \dots, a_{k-1}) \longmapsto$ uniquely determines $\{a_n\}_{n \geq 0}$
 $\uparrow$
 proved by induction.
 $\bigcap_{V_\infty^k}$

It is also surjective

as each sequence $\{a_n\}_{n \geq 0} \in V_\infty^k$

is an image of $(a_0, a_1, \dots, a_{k-1})$.

Thus φ is a linear isomorphism

$\Downarrow$ hence

$$\dim(V^k) = \dim(V_\infty^k)$$

$$\begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{k-1} \end{pmatrix} = a_0 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + a_1 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + a_{k-1} \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \leftarrow \text{canonical basis in } V^k.$$

ALGORITHMS & COMPLEXITY

LECTURE 5

$$\dim(V_{\infty}^k) = k$$

and $a_0, \dots, a_{k-1}$

Thus any scheme satisfying $a_n = \sum_{i=1}^k d_i a_{n-i}$ is a linear combination (so it depends on k -parameters) of the basis sequences $\{e_n^i\}_{n \geq 0}$ satisfying $1 \leq i \leq k$ and $a_n = \sum_{i=1}^k d_i a_{n-i}$ and $a_0, \dots, a_{k-1}$

We shall look at that for different k .

① $\boxed{k=2}$ (or $k=1$)

$$\boxed{a_n = a a_{n-1} + b a_{n-2}}^* \quad \text{linear homogeneous of order 1, 2}$$

(i) take $\underline{b=0}$ & $a \neq 0 \Rightarrow \underline{k=1}$

$$a_n = a a_{n-1}$$

$$\boxed{\dim(V_{\infty}^1) = 1}$$

$\frac{a_n}{a_{n-1}} = a$ so geometric sequence

$$\boxed{a_n = A \cdot a^n}$$

basis 1 sequence

if a_0 is given $\Rightarrow a_0 = a^0 \cdot A \Rightarrow A = a_0$

$\Rightarrow$

unique sequence

$$\boxed{a_n = a_0 a^n}$$

ALGORITHMS & COMPLEXITY

LECTURE 5

(ii) Assume now $a=0$ & $b \neq 0$

$$a_n = b a_{n-2}$$

if a_0 given
 a_1 given

$$\frac{a_n}{a_{n-2}} = b$$

$$a_{2k} = b^k a_0$$

$$a_{2k+1} = b^k a_1$$

Example: (i) $S_n = \underset{a}{3} S_{n-1}$ $S_0 = 5$

$$\boxed{S_n = 5 \cdot 3^n}$$

(ii) $S_n = 3 S_{n-2}$

$$S_0 = 5$$

$$S_1 = 2$$

$$S_{2n} = 5 \cdot 3^n$$

$$S_{2n+1} = 2 \cdot 3^n$$

Case $k=2$ $a \neq 0$ & $b \neq 0$ (iii)

We search for the solution of the form:

$$\boxed{S_n = C \cdot r^n}$$

Plugging into *

$$C r^n = a C r^{n-1} + b C r^{n-2} \quad | : C$$

$$\boxed{r^2 - a r - b = 0} \quad (A)$$

characteristic equation for

$$\boxed{a_n = a a_{n-1} + b a_{n-2}} \quad (AA)$$

ALGORITHMS & COMPLEXITY

LECTURE 5

a) Assume $\Delta > 0$

$\exists \pi_1 \neq \pi_2$ (real numbers) solving Δ

$$\Rightarrow \begin{cases} \{X_n\}_{n \geq 0} = \{\tilde{C}_1 \pi_1^n\}_{n \geq 0} \\ \{Y_n\}_{n \geq 0} = \{\tilde{C}_2 \pi_2^n\}_{n \geq 0} \end{cases}$$

$\tilde{C}_1$ arbitrary
 $\tilde{C}_2$ arbitrary

These two sequences satisfy:

$$S_n = a S_{n-1} + b S_{n-2}$$

It is easy to check that they are linearly independent (and as $\dim(V_\infty^k) = 2$) the form the basis for V_∞^k and thus any solution to (AA) is a linear combination of these basis vectors



$$S_n = \tilde{C}_1 \pi_1^n + \tilde{C}_2 \pi_2^n$$

S_n depends on 2 parameters

if we add S_0 & S_1 we have unique solution S_n

$$\begin{pmatrix} * \\ * \\ * \end{pmatrix} \begin{cases} S_0 = \tilde{C}_1 \pi_1^0 + \tilde{C}_2 \pi_2^0 \\ S_1 = \tilde{C}_1 \pi_1^1 + \tilde{C}_2 \pi_2^1 \end{cases} \Rightarrow \begin{cases} \text{we eliminate } \tilde{C}_1 \text{ \& } \tilde{C}_2 \text{ by} \\ \text{solving } \begin{pmatrix} * \\ * \\ * \end{pmatrix} \\ \text{as we know } S_0, S_1 \end{cases}$$

$\det \begin{pmatrix} 1 & 1 \\ \pi_1 & \pi_2 \end{pmatrix} \neq 0$

ALGORITHMS & COMPLEXITY

LECTURE 5

To see that $\{\tau_1^n\}_{n \geq 0}$ & $\{\tau_2^n\}_{n \geq 0}$

are linearly independent take linear combination:

$$\alpha \{\tau_1^n\}_{n \geq 0} \oplus \beta \{\tau_2^n\}_{n \geq 0} = \{\ominus\}_{n \geq 0} \\ (0, 0, \dots, 0, \dots)$$

$$\Downarrow \\ \alpha \tau_1^n + \beta \tau_2^n = 0 \quad \forall n \in \mathbb{N}$$

take e.g. $n=0$
 $n=1$

$$\textcircled{A} \begin{cases} \alpha + \beta = 0 \\ \tau_1 \alpha + \tau_2 \beta = 0 \end{cases} \Leftrightarrow \begin{bmatrix} 1 & 1 \\ \tau_1 & \tau_2 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\det \begin{bmatrix} 1 & 1 \\ \tau_1 & \tau_2 \end{bmatrix} = \tau_2 - \tau_1 \neq 0 \quad \text{as } \tau_2 \neq \tau_1 \\ \text{as } \Delta > 0$$

so we have one solution to (A)

$$\alpha = \beta = 0 \Rightarrow \{\tau_1^n\}_{n \geq 0} \text{ & } \{\tau_2^n\}_{n \geq 0}$$

are linearly independent.

b) Assume now $\Delta = 0$

$$\Rightarrow \tau_1 = \tau_2 = \tau \quad (\text{multiplicity of } \tau \text{ is two})$$

$$\text{this time } \{\tau_1^n\}_{n \geq 0} \text{ & } \{\tau_2^n\}_{n \geq 0}$$

are linearly dependent !!! so we need to find one more solution to (A) which is

ALGORITHMS & COMPLEXITY

LECTURE 5

not dependent linearly from r^n

$$\{x_n\}_{n \geq 0} = \{C_1 r^n\}_{n \geq 0}$$

take now

$$\{y_n\}_{n \geq 0} = \{C_2 n r^n\}_{n \geq 0}$$

We check:

a) — $\{y_n\}_{n \geq 0}$ satisfies $a_n = a a_{n-1} + b a_{n-2}$

b) — $\{y_n\}_{n \geq 0}$ & $\{x_n\}_{n \geq 0}$ are linearly independent



Any solution to $a_n = a a_{n-1} + b a_{n-2}$
(as $\dim(V_\infty^k) = 2$) is a linear combination:

$$S_n = \tilde{C}_1 r^n + \tilde{C}_2 n r^n \quad (\text{depends on 2 parameters})$$

if we add $\tilde{C}_1$ & $\tilde{C}_2$ as S_0 & S_1 we can eliminate follows:

ALGORITHMS & COMPLEXITY

LECTURE 5

$$* \begin{cases} S_0 = \tilde{C}_1 \tau^0 + \tilde{C}_2 0 \cdot \tau^0 \\ S_1 = \tilde{C}_1 \tau^1 + \tilde{C}_2 1 \cdot \tau^1 \end{cases}$$

$$\det \begin{pmatrix} 1 & 0 \\ \tau & \tau \end{pmatrix} = \tau \neq 0 \quad \text{dla } \tau \neq 0$$

if $\tau = 0$
we have

So we compute

$\tilde{C}_1$ & $\tilde{C}_2$ from (*)

but we assumed $\Rightarrow -b=0$
 $b \neq 0$.

and we get a unique solution to

$$\begin{cases} a_n = a a_{n-1} + b a_{n-2} \\ a_0, a_1 \text{ given} \end{cases}$$

We show a) $r r^n$ satisfies
 $a_n = a a_{n-1} + b a_{n-2}$ (*)
provided
 r^n satisfies (*)

Note normalne charakterystyczne

$$* \quad r^2 - ar - b = 0 \quad \text{if } \Delta = 0 \quad (r - r_1)^2 = 0 \quad **$$

$r_1 \leftarrow \text{root} \quad * \Leftrightarrow **$

$$r^2 - ar - b = 0 \quad \Leftrightarrow \quad r^2 - 2r r_1 + r_1^2 = 0$$

$$\Rightarrow \boxed{a = 2r_1} \quad \boxed{b = -r_1^2}$$

ALGORITHMS & COMPLEXITY

LECTURE 5

Now we put $n\tau_1^m$ into $a_n = a_{n-1} + b_{n-2}$

$$n\tau_1^n = a(n-1)\tau_1^{n-1} + b(n-2)\tau_1^{n-2}$$

$$n\tau_1^n = 2\tau_1(n-1)\tau_1^{n-1} + (-\tau_1^2)(n-2)\tau_1^{n-2}$$

$$n\tau_1^n = 2(n-1)\tau_1^n - (n-2)\tau_1^n \quad | : \tau_1^n$$

$$n = 2n - 2 - n + 2$$

$$n = n$$

$$0 = 0. \quad \text{for a) } \underline{\underline{OK}}$$

b) $\{\tau_1^n\}_{n \geq 0}$ & $\{n\tau_1^n\}_{n \geq 0}$

are linearly independent

$$\alpha \odot \{\tau_1^n\}_{n \geq 0} \oplus \beta \{n\tau_1^n\}_{n \geq 1} = \ominus$$

$$\Downarrow \forall n \geq 0 \quad \alpha \tau_1^n + \beta n \tau_1^n = 0$$

$$\text{take } n=0 \Rightarrow \alpha = 0$$

$$\Rightarrow \beta = 0$$

so $\{\tau_1^n\}_{n \geq 0}$ & $\{n\tau_1^n\}_{n \geq 0}$ are

linearly independent.

ALGORITHMS & COMPLEXITY

LECTURE 5

c) Assume $\Delta < 0$

This is the same as $\Delta > 0$
since we get $r_1 \neq r_2 \in \mathbb{C}$
complex

Example:

a) $a_n = a_{n-1} + 2a_{n-2}$ (*)

$$a_n = C r^n$$

$$C r^n = C r^{n-1} + 2 C r^{n-2}$$

$$r^2 = r + 2 \Rightarrow \underline{r^2 - r - 2 = 0}$$

characteristic equation

$$\Delta > 0 \quad \& \quad \underset{\substack{\parallel \\ 2}}{r_1} \neq r_2 = -1$$

So we have 2 linearly independent solutions 2^n & $(-1)^n$ so

$$C_n = C_1 (-1)^n + C_2 2^n$$

↑ arbitrary solution to (*)

If we add: $a_0 = a_1 = 3$

We can eliminate parameters C_1 & C_2

ALGORITHM & COMPLEXITY

$$\begin{cases} 3 = a_0 = C_1(-1)^0 + C_2 2^0 \\ 3 = a_1 = C_1(-1)^1 + C_2 2^1 \end{cases}$$

$$\hat{=} \begin{cases} 3 = C_1 + C_2 & C_1 = 1 \\ 3 = -C_1 + 2C_2 & C_2 = 2 \end{cases}$$

$$\Rightarrow \boxed{a_m = 2^{m+1} + (-1)^m} \Rightarrow \boxed{a_n = \Theta(2^n)}$$

b) $\boxed{S_m = 6S_{m-1} - 9S_{m-2}} \quad (*)$

characteristic polynomial

$$r^2 - 6r - 9 = 0 \Rightarrow \Delta = 0 \Rightarrow r = 3$$

$$a_m = 3^m \quad \& \quad b_m = m 3^m$$

↖ ↗
basis vectors

Arbitrary solutions to $(*)$ is of the form:
 $S_m = C_1 3^m + C_2 m 3^m$

If initial conditions are known

$\boxed{S_0 = 1 \quad \& \quad S_1 = -3}$ then
we can eliminate C_1 & C_2

$$\begin{cases} 1 = S_0 = C_1 3^0 + C_2 \cdot 0 \cdot 3^0 \\ -3 = S_1 = C_1 3^1 + C_2 1 \cdot 3^1 \end{cases}$$

ALGORITHMS & COMPLEXITY

LECTURE 5

$\Rightarrow c_1 = 1$ & $c_2 = -2$ & thus

$$\boxed{S_n = 3^n - 2 \cdot n \cdot 3^n} \Rightarrow \boxed{S_n = \Theta(n 3^n)}$$

a unique sequence satisfying
(*) & $S_0 = 1$ & $S_1 = -3$

The cases for $K > 2$ can be analyzed similarly but the multiplicities or roots of characteristic polynomial decide about the basis for V_∞^K .

Remark: Once you find a_n in the explicit form (whatever method you use) check whether your solution satisfies recursive scheme & initial conditions to eliminate the arithmetic error which potentially may happen in your calculations!!!